

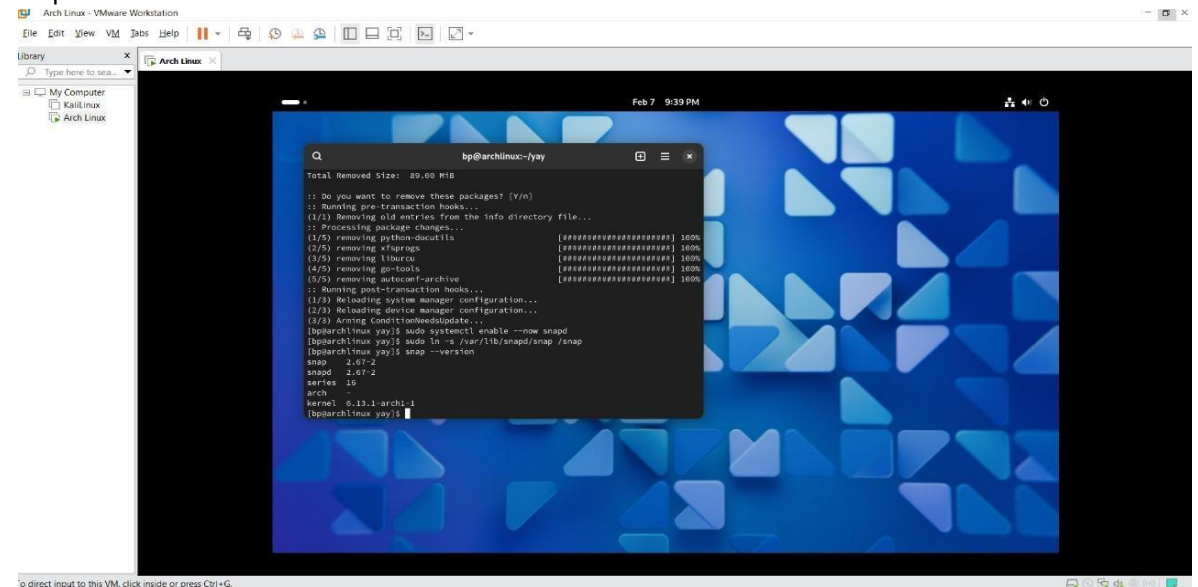
Task 2: Understanding Snap, DEB, and GTK+ Packages

1. Introduction

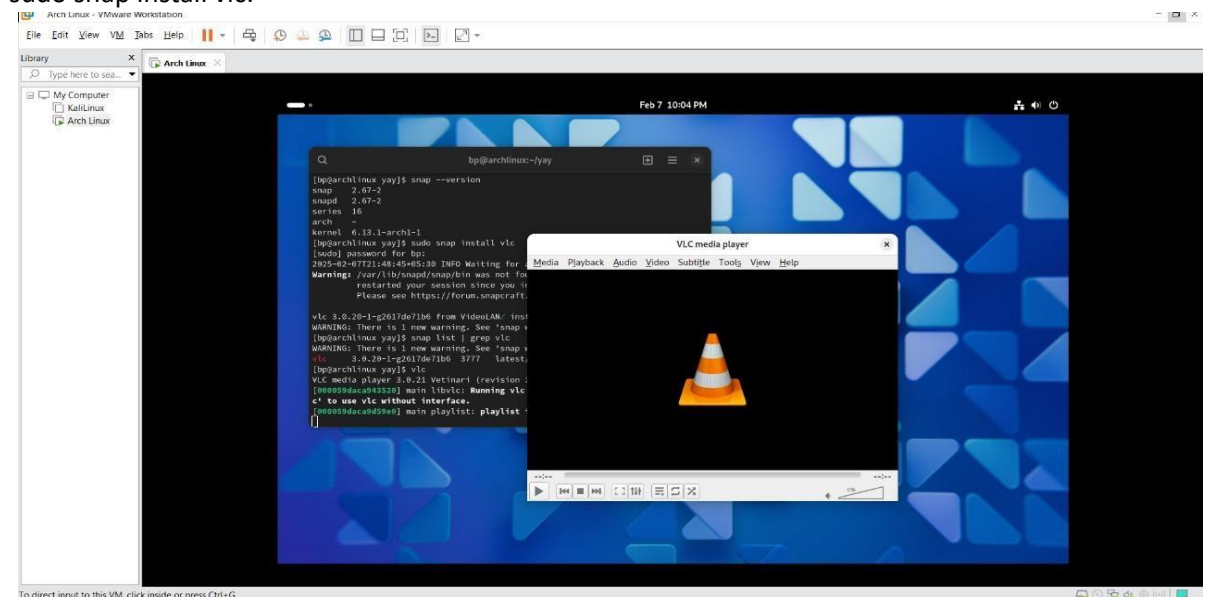
- **Snap packages** offer a universal, containerized format designed for cross-distribution compatibility, bundling all dependencies and managed by snapd.
- **DEB packages** are the standard software format for Debian-based Linux distributions, relying on shared system libraries and managed by APT.
- **GTK+** is a toolkit used for building graphical user interfaces, employed by applications packaged in both DEB and Snap formats.

Snap Package Installation:

Snap installation:



sudo snap install vlc:



Steps:

- Download the .deb File
- Convert the .deb Package to an Arch-Compatible Format using debtool command because **direct use of dpkg crashes the system**
- Install the Converted Package Using pacman
- As a proof of successful Installation, **install chrome**

The screenshot shows the Arch Linux VM installation progress in VMware Workstation. The main window displays the Arch Linux desktop environment with a blue geometric pattern. A terminal window is open, showing the installation progress of the 'google-chrome' package. The terminal output includes the following commands and messages:

```

bp@archlinux:~/Downloads
Saving to: '/home/bp/Downloads/google-chrome.deb'
/home/bp/Downloads/ 100%[=====] 108.79M  862KB/s   in 5s 3s

2025-02-07 23:50:36 (367 KB/s) - '/home/bp/Downloads/google-chrome.deb' saved [1
14071328/114071328]

[bp@archlinux Downloads]$ ls -l ~/Downloads/google-chrome.deb
-rw-r--r-- 1 bp bp 114071328 Feb  4 23:22 /home/bp/Downloads/google-chrome.deb
[bp@archlinux Downloads]$ cd ~/Downloads
[bp@archlinux Downloads]$ dpkg-deb -x google-chrome.deb
==> Extracting package data...
==> Fixing possible directories structure differences...
==> Generating .PKGINFO file...

?: Enter Package name (can be left blank):

?: Enter package license (can be left blank, you can enter multiple licenses com
ma separated):

... Creation of .PKGINFO file in progress. It may take a few minutes, please wai
t...
```

The terminal window also shows the file size of the downloaded package (108.79M) and the time taken to download it (5s 3s). The background of the VM desktop is a blue geometric pattern.

The screenshot shows a desktop environment with a blue geometric wallpaper. In the foreground, a Google Chrome window is open, displaying the 'Sign in to Chrome' screen. The screen features the Chrome logo and the text 'Sign in to Chrome' and 'Get additional features'. Below this, there are three cards: 'All your devices' (with a laptop icon), 'Stronger security' (with a shield icon), and 'Back up data' (with a cloud icon). At the bottom right, there are two buttons: 'Don't sign in' and 'Sign in'. In the background, a terminal window is open, showing the command prompt 'bp@archlinux:~/Downloads' and the command 'google-chrome'. The terminal output shows the command being executed and the browser window opening. The desktop environment includes a taskbar at the top with various icons and a sidebar on the left with a file manager and application launcher.

GTK+ application

Gtk4 installation:

